

NIKIL KRISHNA *Graduate Student*

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ABOUT ME

My research lie within structural computational biology and computational drug discovery. My work spans molecular modeling, virtual screening, molecular dynamics, protein engineering, and scientific computing, with hands-on experience developing and automating computational workflows in Python. I am particularly interested and learning more about applying machine learning and generative models to protein structure prediction, drug discovery, and protein binder design.

EXPERIENCE

GATE Biotechnology 04/2026

- AIR 14
- Score 858

Capstone Project and Publication 04/2026

SASTRA Deemed University

- Structure-Based Drug Design of Novel Inhibitors Targeting the Thiamine Biosynthesis Pathway in Bacteria
- High throughput **Virtual Screening**
- **Lead Optimization** by R-Group substitution and SAR analysis
- Molecular Dynamics and Analysis
- **Free Energy Perturbation** to find Absolute Free Binding Energy
- Submitted to GEN Biotechnology journal.

Summer Internship 06/2025

IIT Dharwad

- Development of Early-Stage Computational **Drug Discovery Pipeline** Using **Python**
- Virtual Screening, Rescoring and **Statistical Analysis**
- Virtual Environment and dependency management using **Conda**
- Parallel High Performance **Scientific Computing**
- Benchmark of three SOTA **machine learning** architectures for protein binder design

Undergraduate Student Researcher 06/2024

SASTRA Deemed University

- Molecular Motors Laboratory
- Drug Design and Protein Engineering
- Research Credits - Fragment Based Drug Design Of Anthelmintic using Genetic Algorithm

EDUCATION

B.tech Biotechnology 2022

SASTRA Deemed University, Thanjavur

HSC and SSLC 2020

ALPHA Wisdom Vidyashram

SKILLS

ML Protein Structure Prediction

- AlphaFold
- Boltz
- Diffusion Based Models

Drug Design

- Virtual Screening
- Fragment Based Drug Design
- Genetic algorithm/Combinatorial Chemistry
- Lead Optimization
- ADMET analysis

Molecular Docking and Dynamics

- Autodock
- DiffeDock
- Gromacs
- OpenMM

Process Simulation

- Aspen Plus
- SuperPro Designer
- DWSIM

Molecular Modelling

- ChimeraX
- PyMOL
- VMD
- MOE
- Schrodinger Maestro

Python

- Molecular Visualization
- Data Analysis
- Machine Learning
- Bioinformatics Pipeline construction
- Dependency management
- Scripting and Automation

Bioinformatics

- Linux
- Visual Studio Code
- Git version management
- Scientific Computing
- Scientific writing

LANGUAGES

English

Professional working proficiency

Hindi

Limited working proficiency

Tamil

Bilingual proficiency

Kannada

Bilingual proficiency

Telugu

Elementary proficiency

CERTIFICATIONS

NPTEL

Aspen Plus Simulation
Software - A Basic Course
For Beginners

- 90%

Schrodinger Drug

Discovery Hackathon

Winner

Insilco Structure-based Drug
Design targeting
Mycobacterium Tuberculosis
using genetic algorithm and
combinatorial chemistry.

NPTEL

2024 Drug Delivery:
Principles and Engineering

- 83%